

Climate, Migration and Regional Development

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April 2026

Abstract

This paper investigates migration as it relates to both climate and regional development. Leveraging the reduction in travel costs induced by the construction of Brasília, a planned capital city, I show that people in regions exposed to extreme climates do not respond to opportunities to leave at greater levels than otherwise. Then, using the migration flows induced by reduced costs, I show that net migrant receipt induces higher welfare, as measured by growth in both GDP and GDP per capita. These flows are estimated to have led to an increase of 2.3% in aggregate Brazilian GDP per capita from 1970 to 2010.

1 Introduction

Being born is the single most crucial lottery we ever play, and might be the most random one. Many decisions in life can be seen as attempts to attenuate this one roll of the dice, and migration is one of the more effective escape routes from bad rolls. In particular, being born in a place recurrently exposed to flooding, heatwaves or droughts may yield a lifelong disamenity, from which the only alleviation is getting out; and those who stay in such places may benefit from these out migrations, as more resources have to be distributed among fewer people, or suffer from them, as their most productive compatriots leave to seek a better life elsewhere. The same logic applies to poorer regions, and thus migration might be a relevant driver either convergence or inequality between regions.

This paper is an attempt to bring clarity to these issues. At first, I ask whether people exposed to bad climate are more likely to explore migration-facilitating technologies in the form of lower travel costs; then I explore migration's effect on a real life proxy for welfare. There are two main contributions: I find no evidence that those exposed to worse climate respond more strongly to the opportunity to leave; and show in a model free environment that migration has significant impacts on local welfare and through this channel, is a driver of both aggregate welfare gains and regional inequality in how these gains are distributed.

Whenever studying migration's impacts on welfare endogeneity is an obvious concern. Even if receiving migrants may seem to induce higher welfare, it is immediate that the reverse is also true: high welfare regions ought to attract more migrants, as do regions where welfare is increasing rapidly. Simple correlations and regressions will inevitably be biased and not too informative. The same is true for the lowering of travel times, as a new road may induce larger migratory flows but, likewise, governments may choose to build roads in response to anticipated migration, or in migratory routes already under use. Codetermination is pervasive in this area of study.

RCTs being impossible in such a setting, the surest way of disentangling these effects is to find an instrument that impacts one of the channels of interest but plausibly has nothing directly to do with the outcome. Luckily, the Brazilian experience in the XXth century provides us with just such a quasi experiment: the construction of Brasília, the country's planned capital city. Brasília was the brainchild of Juscelino Kubistchek, and its construction led to a large drive of road building through the country, as it, located in the geographical center of Brazil, had to be connected to the densely populated coastal areas. This boom allows for just the kind of instrument we would wish for: a clean lowering of travel times that has nothing to do with previous migration; and through it, an increase in migratory flows that has nothing to do with previous welfare outcomes. In obtaining the instrument I

follow Morten and Oliveira (2024).

With exogenous variation in travel times in hands, estimating the interaction of climate to migration response intensity is straightforward. Using granular data on Brazilian climate conditions, I create indexes for extreme climate exposure, which I then interact with the instrumented travel time reductions. As Brazilian census data allows recovery of migration flows, not simply population variations, this can then be used directly to test for the relevance of the interaction term. I find no evidence of relevance, independently of the climate exposure measure used.

The causal effect of migration on welfare remains more elusive. Exogenous travel time reductions allows for obtaining of plausibly exogenous migration flows, but the reductions themselves also impact GDP per capita through other channels, such as increased trade. To account for that, I construct gravity model implied controls. This allows me to identify the isolated effect of migration on GDP per capita growth, which is positive and significant, as well as robust to different measurements. Finally, combining the migration effects on each region of the country allows for extracting the aggregate contribution of migration to GDP per capita, which sits at 2.3%.

Two extensions suggest themselves. First, building better controls for the other effects that travel time reductions have on welfare would be desirable. Data limits the richness of such controls, but gains can be made by shifting the analysis unit from mesorregions, for which we have only GDP and population data, to state levels, for which we have actual trade flows. Second, investigating heterogeneity in the type of migrants each regions is receiving could shed light on the mechanisms behind the increase in GDP per capita.

2 Related Literature

This work is related to two broad branches of the economic literature on migration. Many studies have been conducted on the relations of climate and migratory flows, with ambiguous results. Focusing on disasters, Stanley (2010) and Hanson & McIntosh (2010) find positive effects, but Findley (1994) sees no permanent migration outcomes. I add one point in favor of the later, highlight how context dependent these outcomes are.

The relation of migration and welfare has also been widely investigated, in both structural and reduced form contexts. On the first approach, Desmet & Rossi-Hansberg (2021) provide evidence of how migration mitigates welfare losses due to locally heterogenous shocks, and Morten & Bryan (2019) explore de Indonesian setting and a spatial model to indicate how reducing migration barriers could deliver high productivity gains. On the second, Bryan & Mobarak (2014) document how financing temporary migration in Bangladesh delivers

substantial welfare gains to families affected by hunger. My main contribution to this strand of the literature is delivering evidence of migration’s effect on a direct measure of welfare.

This paper is heavily reliant on work done by Oliveira & Morten (2024). I build on their foundation of using Brasília as an instrument to road-building and rely on the dataset they built for roads in Brazil. At this stage I also draw on their compilation of census data. I am deeply indebted to them.

3 Data

The empirical strategy combines four main data sources, each contributing a different dimension of the climate–migration–development nexus.

Migration flows and transport costs are taken from Morten & Oliveira (2025). Migration is constructed from Brazilian census data, yielding bilateral flows at the mesoregion, of which there are 137, level for 1980–2010 at decennial frequency. Travel time between regions over the 1940–2010 period was obtained by compiling geocoded data from Brazilian government sources on paved roads and running a fast-marching algorithm through them. Roadless travel time, used as baseline, was obtained by running the same algorithm on the no-roads map.

Bilateral mesoregional migration flows are sparse: in most years, most pairs record zero flows. It is also top-heavy, with few mesoregions receiving large shares of total migrants. Finally, it is spatially correlated, as nearby mesoregions display similar dynamics.

Climate conditions are derived from the BR-DWGD dataset, which provides daily weather realizations from the 1960s onward. These data are aggregated into mesoregion-level exposure measures. I first produce counts of extreme event instances for each of heatwaves (which trigger when daily median temperature is more than 4.5C above the summer average), flooding (triggering if a day sees over 55mm of rain), drought anomalies (computed according to a standard SEPI index) and extreme relative drought (triggering when a given mesoregion is over 1.5 standard deviations more arid than the rest of the country). From these, I construct a composite indicator of local extreme climate exposure by taking means of Z scores, squaring the result and averaging over a decade.

Economic outcomes are measured using GDP and population data from IPEA, available at decennial frequency from 1940 to 2010. Our main outcome is GDP per capita, which we interpret as a model-agnostic proxy for local welfare.

The combination of these datasets imposes important trade-offs. Climate measures are unavailable for the early (pre-Brasília) period, constraining the joint analysis of climate and migration to the post-1960 decades. Mesoregional migration data is also unavailable for pre

Brasília periods, and trade data is utterly lacking.

Descriptive statistics and regional distribution (provided in Appendix A) show substantial dispersion in all key variables. Spatial patterns are also a key concern. Climate exposure is concentrated in the Northeast (where the "polígono das secas" has been known as a climate vulnerable region for over a century), the Cerrado and the extreme southwest, while migration and GDP per capita growth display marked regional heterogeneity.

4 Identification

The identification strategy proceeds in three steps: I first reproduce the Brasília travel time instrument, which is extended to incorporate climate heterogeneity. The third step is generating exogenous migrations flows to study welfare outcomes.

The core source of exogenous variation comes from the construction of Brasília and the associated expansion of the national road network. Following Morten and Oliveira (2025), we use predicted reductions in bilateral travel time—driven purely by the geography of the new road system—as an instrument for migration costs. This generates plausibly exogenous shocks to migration flows across origin–destination pairs. We first use these shocks to identify migration responses. Specifically, we estimate gravity-style specifications of bilateral migration flows on instrumented travel times, including rich origin–year and destination–year fixed effects. These absorb all time-varying local conditions, so identification comes from differential changes in connectivity across pairs. To assess the role of climate, we interact the instrument with measures of local exposure, asking whether regions facing worse climate conditions respond more strongly to reductions in migration costs.

In a second step, we aggregate the predicted migration flows to the regional level and construct a measure of net migration induced solely by transport shocks. Due to the rich fixed effect setting, this object is by construction orthogonal to contemporaneous local economic conditions and thus provides a source of exogenous variation in migration when studying its effects on welfare.

A key concern is that the underlying shock—road construction—may directly affect economic outcomes beyond migration (e.g., through market integration or trade). To address this, we include a gravity-inspired market access control, capturing the direct effects of improved connectivity on local economic activity. Identification of the welfare channel thus comes from variation in migration that is orthogonal to both local fundamentals (absorbed by fixed effects) and direct transport-induced gains (captured by market access).

5 Results

I begin by replicating the first-stage relationship between the Brasília-induced instrument and travel times, obtaining results consistent with the original paper. Instrumented reductions in travel time lead to large and precisely estimated increases in bilateral migration flows. As this is a mere replication, these results are presented in Appendix B.

I then turn to the interaction with climate exposure. Table 1 shows that the elasticity of migration with respect to travel time is large (around 1.4) and highly significant across specifications. However, the interaction with climate exposure is small and statistically insignificant. This suggests that, while climate conditions may affect migration levels, they do not systematically amplify responsiveness to changes in migration costs. In other words, there is little evidence for the “climate-amplified migration response” channel.

With this elasticity and fixed effects in hand, and in order to produce exogenous migration flows I use the prior to Brasília map of Brazilian roads to generate ex-Brasília counterfactual migration flows. Conflating these to the share of flows explained by the original regression allows for recovery of flows due solely to the reduction in travel times caused by Brasília’s construction.

Finally, for each origin mesorregion-year I build a market access index by dividing destination GDP by the distance (measured in travel time) to it and aggregating destinations. As the construction of Brasília lowers travel times it leads to an increase in market access.

Having both control and treatment variables, I now proceed to the second set of results, which focuses on the welfare effects of migration. Using the instrumented transport-induced migration measure, Table 2 shows that net migrant inflows have a positive and significant effect on both GDP and GDP per capita growth. The estimates are economically meaningful: regions receiving more migrants due to exogenous reductions in migration costs experience faster income growth, with effects robust across random- and fixed-effects specifications. By contrast, the direct market access control is small and statistically insignificant, suggesting either that the estimated effects operate primarily through migration rather than direct access-to-GDP channels or that the chosen measure is inadequate.

Spatial patterns tell a very clean story, as shown on Figure 1. Regions along the agricultural frontier and already urbanized areas emerge as net beneficiaries of migration-induced growth, while historically disadvantaged regions—particularly in the Northeast—experience relative losses, consistent with out-migration and weaker local growth dynamics. Aggregating these effects, internal migration is estimated to have increased Brazilian GDP per capita by approximately 2.3% between 1970 and 2010. Although higher, this value is on the same order of magnitude of the model implied outcome for total welfare obtained by Morten &

Oliveira (2025), which sits at 0.6%

Table 1: Climate Exposure and the Effect of Instrumented Travel Time on Migration

	<i>Dependent variable: Log migration flow</i>		
	Baseline	Climate Exposure	Exposure Dummy
IV time travel	-1.432*** (0.044)	-1.420*** (0.046)	-1.432*** (0.044)
Climate exposure $\times$ IV tt		-0.064 (0.050)	
Exposure Dummy $\times$ IV tt			-0.010 (0.097)
Observations	43,640	43,640	43,640

Notes: Observations are origin–destination–year. Origin $\times$ year and destination $\times$ year fixed effects are included. Standard errors are clustered at the origin–destination level.

Table 2: Migration-Induced Growth Effects

	GDP per capita growth		GDP growth	
	RE	FE	RE	FE
Brasília induced net migrant receival rate	0.046*** (0.011)	0.081*** (0.018)	0.203*** (0.014)	0.224*** (0.023)
Market access gained	0.000 (0.000)		0.000 (0.000)	
Observations	547	547	547	547
Log-likelihood	1271.970	—	1196.057	—

Notes: Dependent variables are annual log growth in GDP per capita and GDP, respectively. Intercepts and year coefficients are omitted. Random-effects (RE) specifications include year fixed effects and random effects for region and region-specific time trends. Fixed-effects (FE) specifications include region fixed effects, year fixed effects, and region-specific time trends. Standard errors in parentheses.

6 Conclusion and Further Work

This draft paper set out to investigate whether climate disamenities amplify migration responses to reductions in migration costs, and whether this mechanism contributes to regional inequality. Results have been twofold.

There is little support for the hypothesis that climate exposure amplifies migration responses. While climate conditions may matter for incital levels of migration, they do not appear to systematically affect how strongly regions respond to changes in migration costs.

On the welfare side, migration flows have strong effects on local economic outcomes: receiving migrants induces faster regional growth in both GDP and GDP per capita terms. Internal migration thus emerges as a driver of both national and regional economic dynamics in Brazil over the second half of the twentieth century. Regionally, even as economic inequality fell migration acted to increase it, creating winners and losers. Nationally, it retained a positive effect on aggregate welfare as measured by GDP per capita. This is evidence that

GDP per Capita Growth due Migration 1970-2010

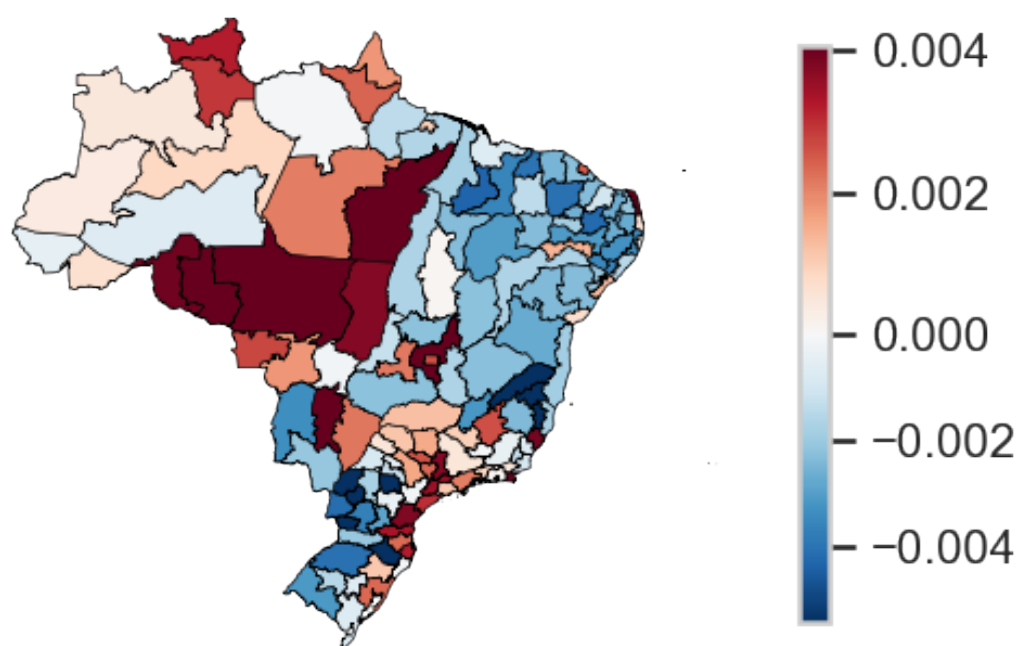


Figure 1: Regional patterns in migration effect on GDP per capita

migration leads to more efficient allocation of labour, rather than simply shifting resources around.

These results are not without caveats. The empirical strategy delivers plausibly exogenous variation in migration, but the underlying shock—road construction—may affect economic outcomes through multiple other channels. Although I control for market access in a reduced-form way, this is unlikely to fully capture either the broader general equilibrium effects of improved connectivity or the direct impacts via trade. A more convincing case would require a replication of the full pipeline at the state level, where longer time series and pairwise trade data are available.

Looking forward, the results open two promising avenues for further work, as the mechanisms behind the positive effect of migration on regional growth remain to be fully understood. The first one is applied, and relates to selection in migration, a phenomenon that has been documented in other settings: people choosing to migrate are probably those who anticipate better outcomes on the place of destination. Whether or not this was the case in the episode studied here could be addressed using census microdata on migrants. The second one is structural: the heterogeneous spatial patterns seen in Figure 1 indicate that both agglomeration in cities and resource endowments in the agricultural frontier play a central role in shaping the relation of migration and welfare outcomes. Formalizing these mechanisms within a spatial equilibrium framework could help rationalize the observed dynamics and connect them to broader theories of development.

Taken together, these extensions would not only strengthen the identification of the core results but also contribute to a deeper understanding of Brazil’s mid- to late-twentieth-century development process.

7 Appendix A

Table 3: Summary Statistics: GDP and Population Growth

Variable	N	Mean	σ	Min	P25	Median	P75	Max	
GDP per capita growth	2,857	0.03	0.06	-0.42	-0.01	0.03	0.06	0.74	[flushleft]
Population growth	2,859	0.02	0.02	-0.13	0.01	0.01	0.02	0.37	
GDP growth	2,857	0.05	0.07	-0.39	0.01	0.05	0.09	0.75	

Notes: Sample covers 1920–2010 for 137 mesoregions (fewer regions available in early decades). All variables are annualized log changes.

Table 4: Summary Statistics: Migration Flows and Travel Time

Variable	N	Mean	σ	Min	P25	Median	P75	Max	
Migration flows	74,528	459	2,304	0	0	24	128	89,896	[flushleft]
Log travel time absent roads	74,528	4.85	0.74	1.27	4.44	5.03	5.41	6.00	
Log travel time on roads	74,528	3.24	0.88	1.03	2.69	3.24	3.71	5.48	

Notes: Sample covers 1980–2010, decennial observations for origin–destination mesoregion pairs. Migration flows are in levels; travel times were obtained via a fast-marching algorithm (Morten & Oliveira, 2025).

Table 5: Summary Statistics: Climate Exposure Indexes

Variable	N	Mean	σ	Min	P25	Median	P75	Max	
Composite exposure	2,466	0.23	0.49	0.00	0.04	0.12	0.21	6.77	[flushleft]
Drought anomaly	2,466	0.00	1.00	-0.79	-0.55	-0.42	0.17	7.77	
Drought absolute	2,466	0.00	1.00	-0.55	-0.47	-0.44	-0.40	5.01	
Heat exposure	2,466	0.00	1.00	-0.54	-0.36	-0.32	-0.22	9.89	
Flood exposure	2,466	0.00	1.00	-0.61	-0.49	-0.40	-0.29	8.24	

Notes: Original data are daily observations from 1961–2024 at $0.1^\circ \times 0.1^\circ$ resolution, including precipitation, temperature (min/max), relative humidity, and evapotranspiration. Variables shown are decennial averages of standardized exposure indexes aggregated to the regional level.

8 Appendix B

Mean Extreme Climate Exposure 1970-2010

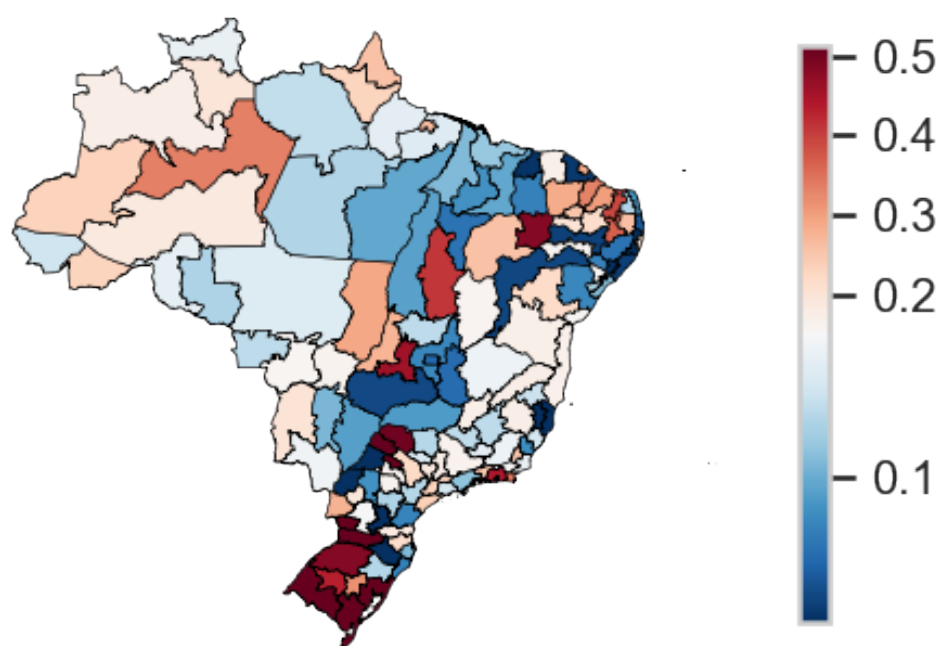


Figure 2: Climate exposure

Net Migrant Receiving Rate 1980-2010, share of population

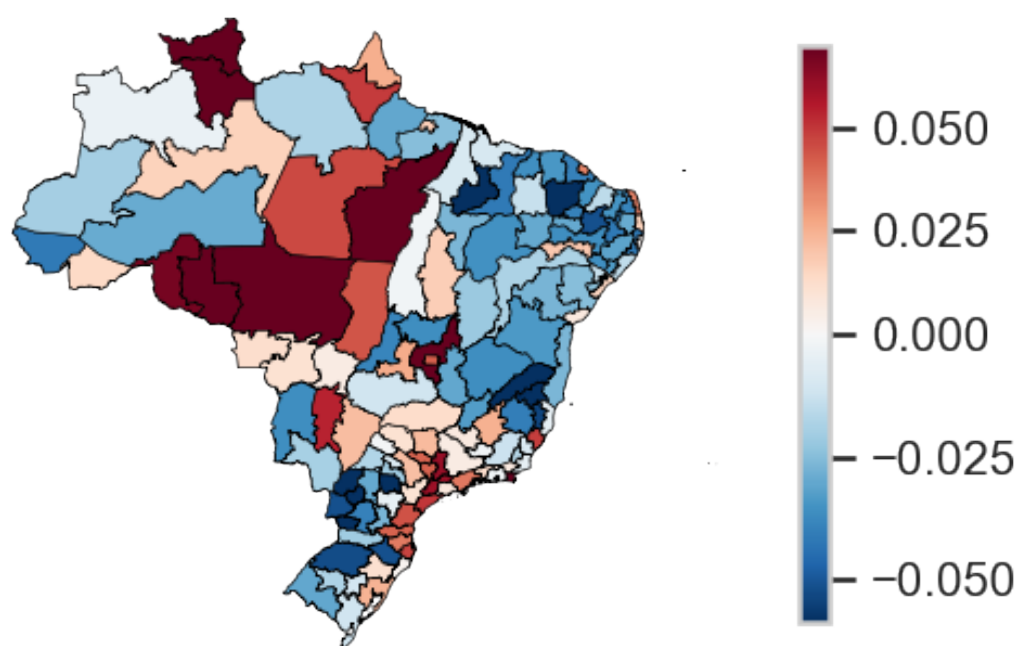


Figure 3: Migration

Annualized GDP per capita Growth Rate 1970-2010

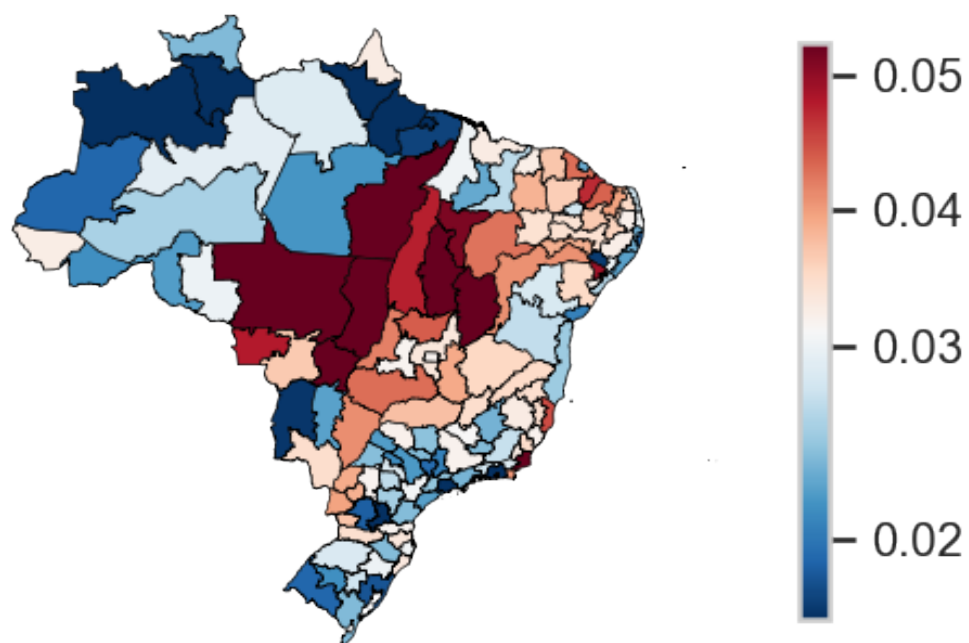


Figure 4: GDP per capita growth

Table 6: Effect of Brasília-Induced Travel Time Reduction on Travel Time on Roads

	<i>Dependent variable: log(fm_road)</i>
Brasília Induced TT Reduction $\times$ 1940	-0.023*** (0.005)
Brasília Induced TT Reduction $\times$ 1960	-0.210*** (0.005)
Brasília Induced TT Reduction $\times$ 1970	-0.378*** (0.005)
Brasília Induced TT Reduction $\times$ 1980	-0.575*** (0.005)
Brasília Induced TT Reduction $\times$ 1990	-0.544*** (0.005)
Brasília Induced TT Reduction $\times$ 2000	-0.545*** (0.005)
Brasília Induced TT Reduction $\times$ 2010	-0.540*** (0.005)
Observations	149,056